

Sparse (BM25)

Query terms



Tokenise + stem



Term-frequency
scoring



Inverted index
lookup



Top- k passages

Dense (mDPR)

Query text



Encoder
→ 768-d vector



Inner-product
with passage vectors



FAISS index
(approx. NN)



Top- k passages

*complementary strengths:
BM25 high recall, mDPR high precision*